

Lecture 7: Basal slip

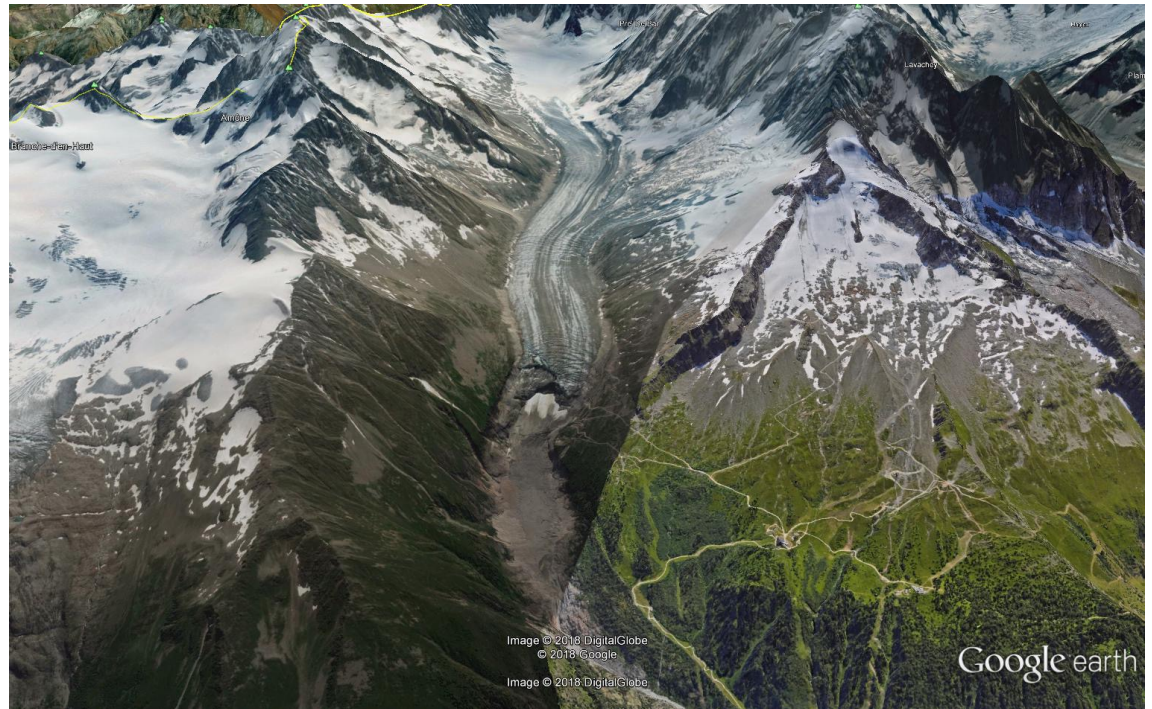
Glaciers slide:

How do we know?

What controls how
fast this happens?

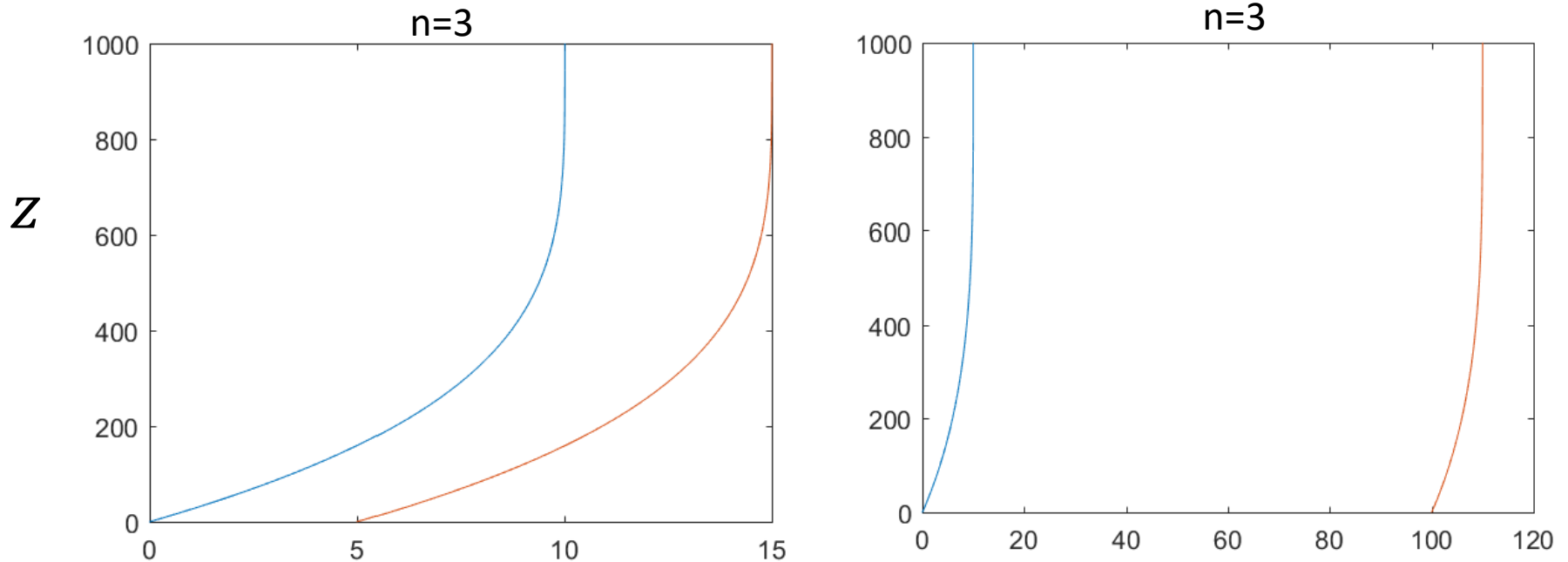
How do we model slip?

Implications?



<https://www.youtube.com/watch?v=njTjfJcAsBg>

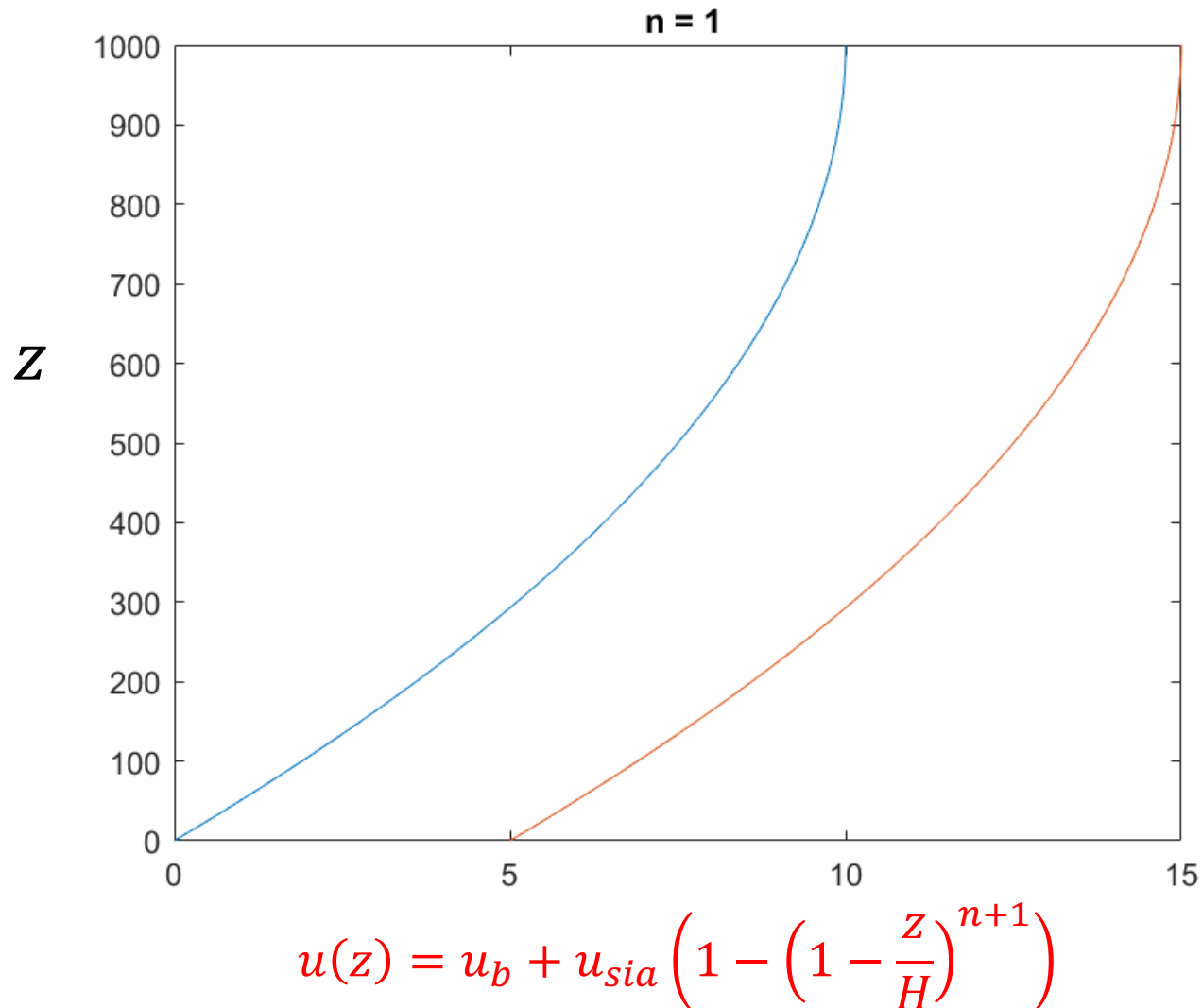
“Sliding” and “basal slip”



$$u(z) = u_{sia} \left(1 - \left(1 - \frac{z}{H} \right)^{n+1} \right)$$

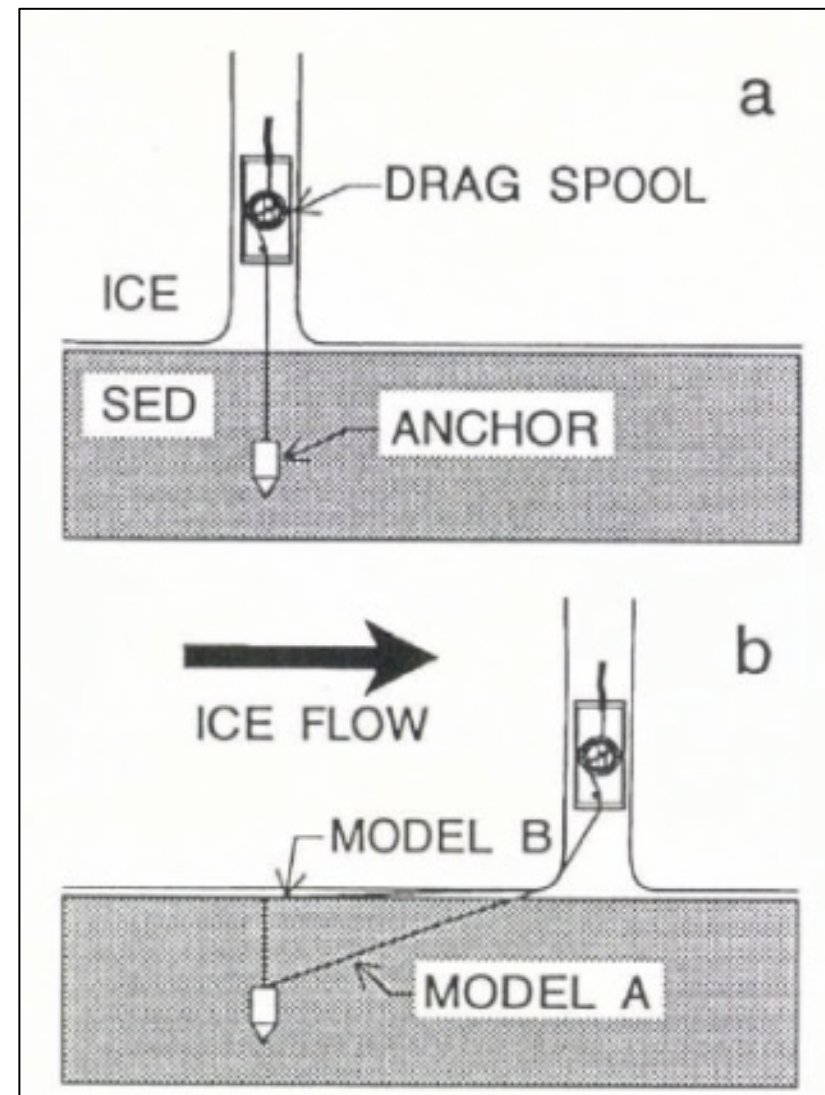
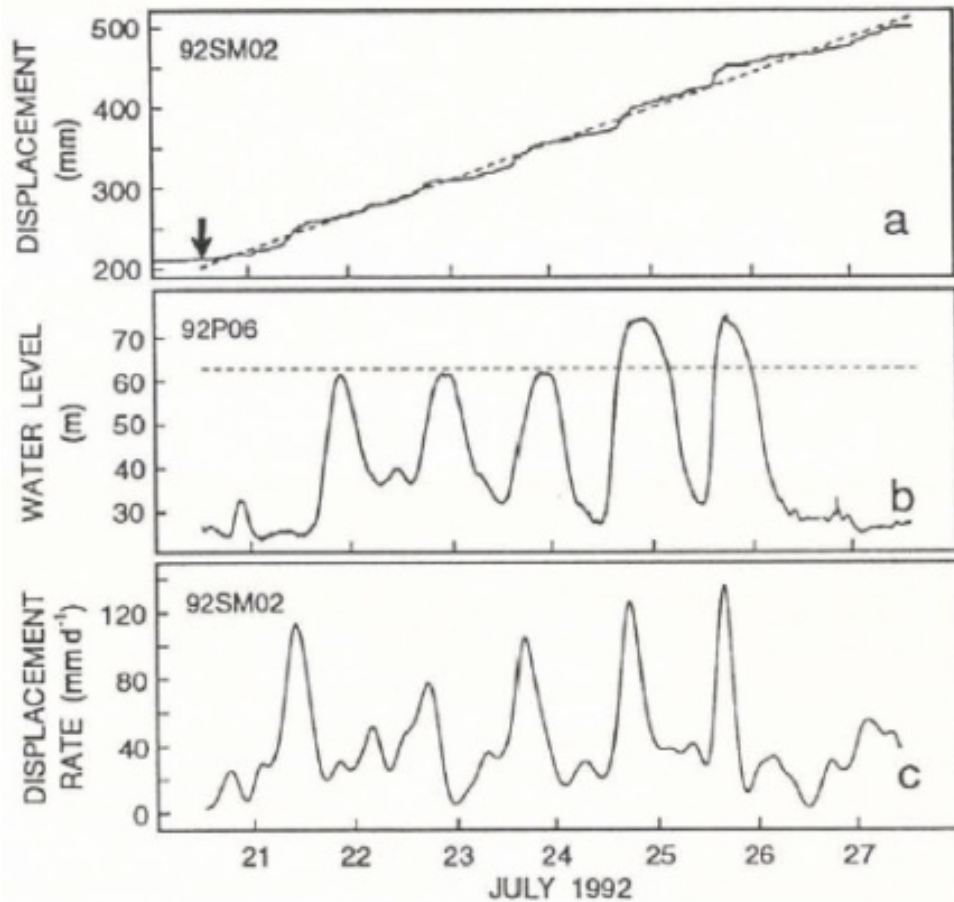
$$u(z) = u_b + u_{sia} \left(1 - \left(1 - \frac{z}{H} \right)^{n+1} \right)$$

“Sliding” and “basal slip”



How do we know?

1. Direct observations



50-70% was slip, the rest was sediment deformation.

How do we know?

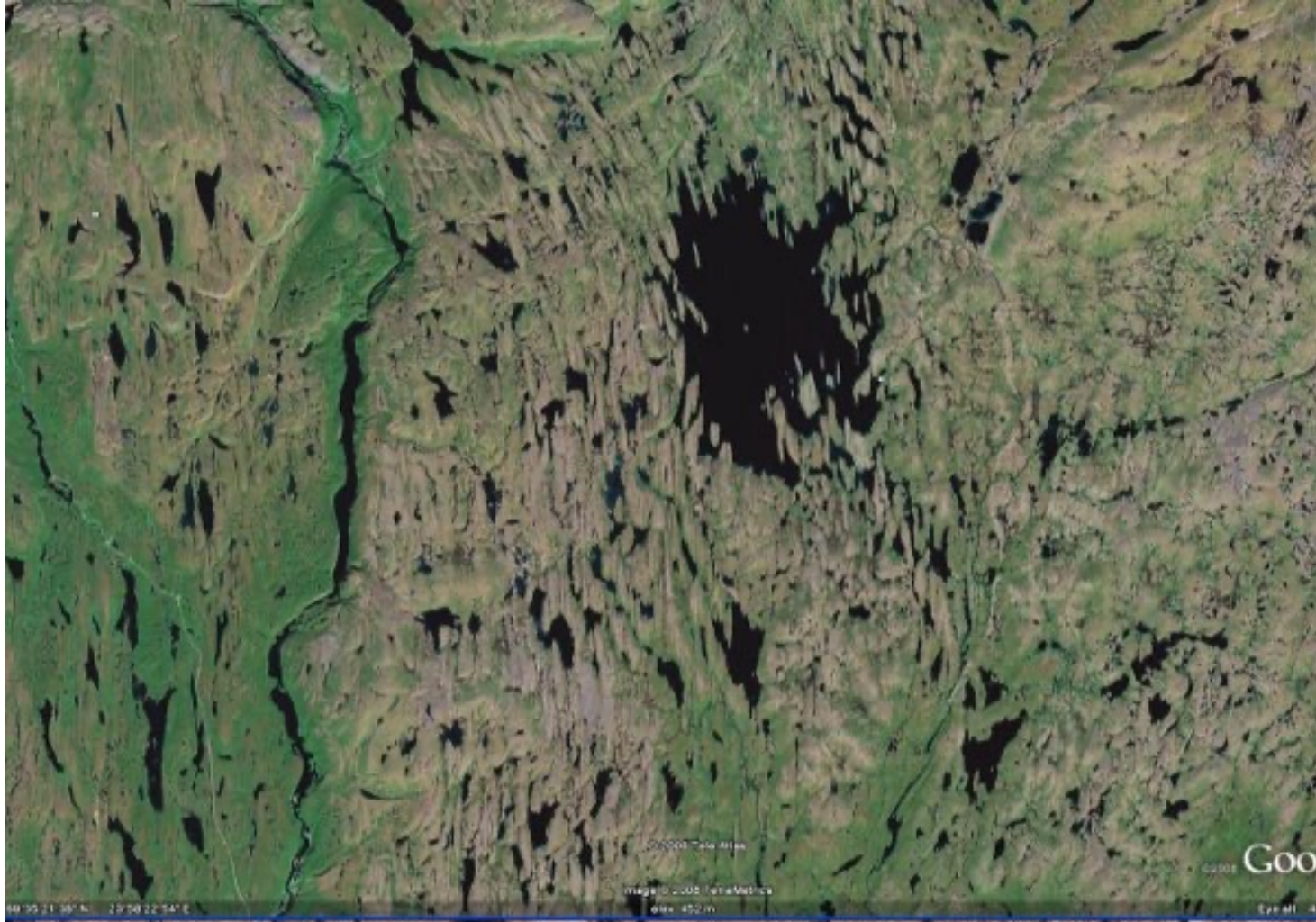
2. Bedforms



Chris Clark's website: <https://www.sheffield.ac.uk/drumlins/drumlins>

How do we know?

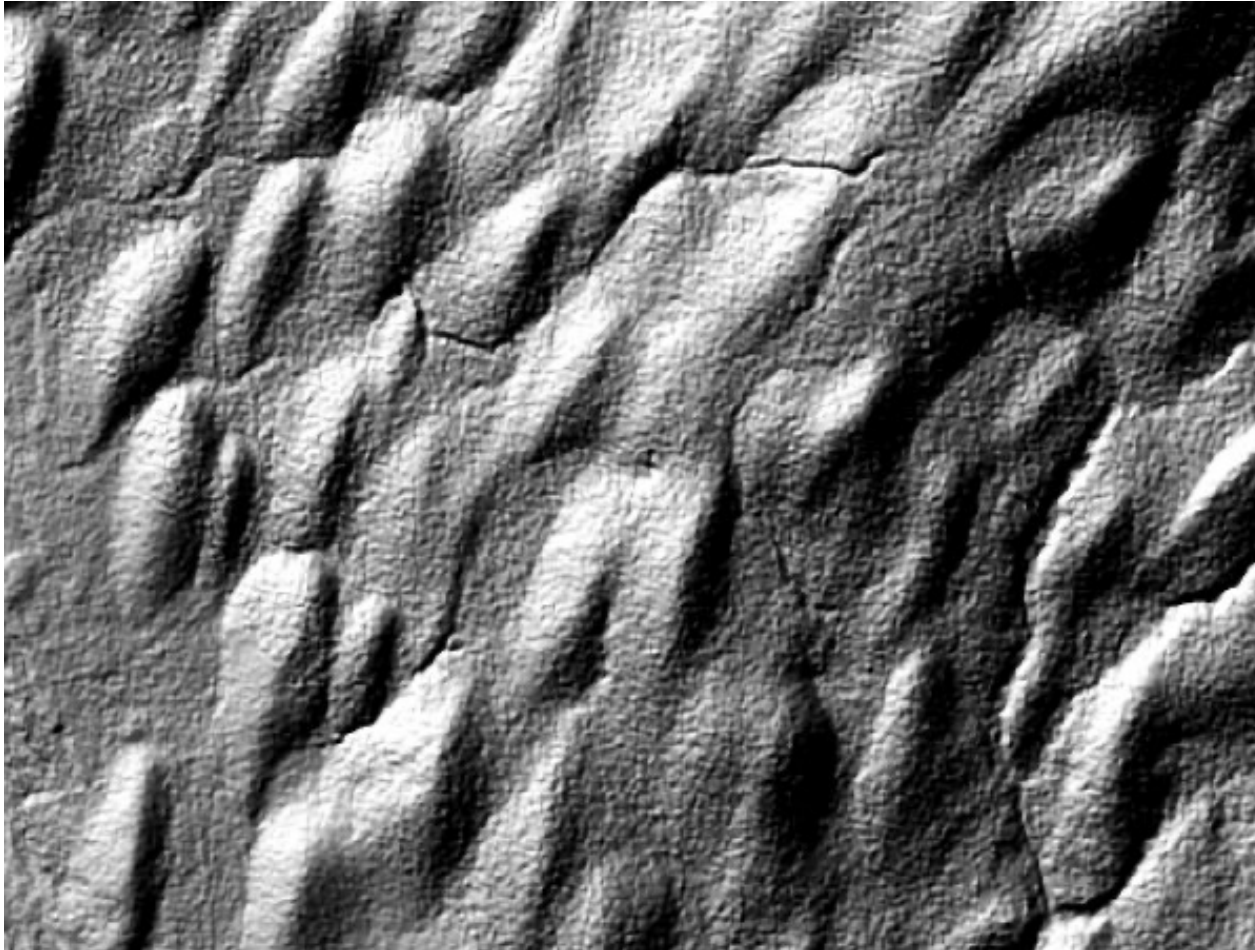
2. Bedforms



Chris Clark's website: <https://www.sheffield.ac.uk/drumlins/drumlins>

How do we know?

2. Bedforms



Chris Clark's website: <https://www.sheffield.ac.uk/drumlins/drumlins>

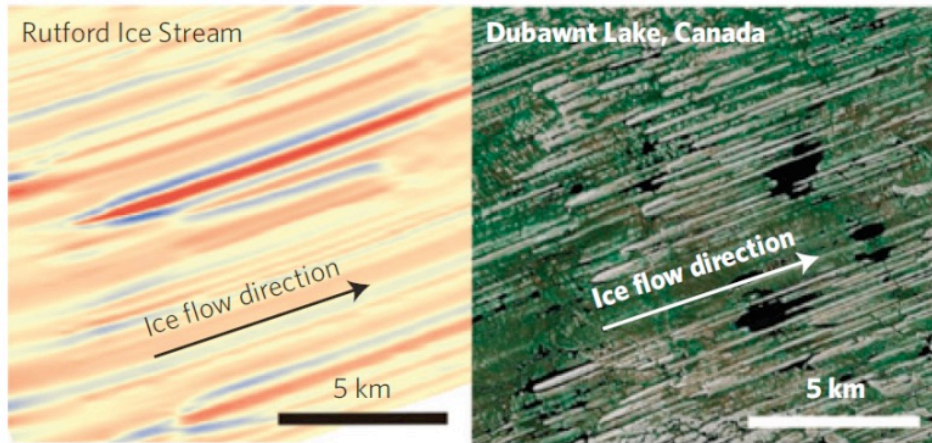
How do we know?

2. Bedforms



How do we know?

2. Bedforms



King et al. (2009)



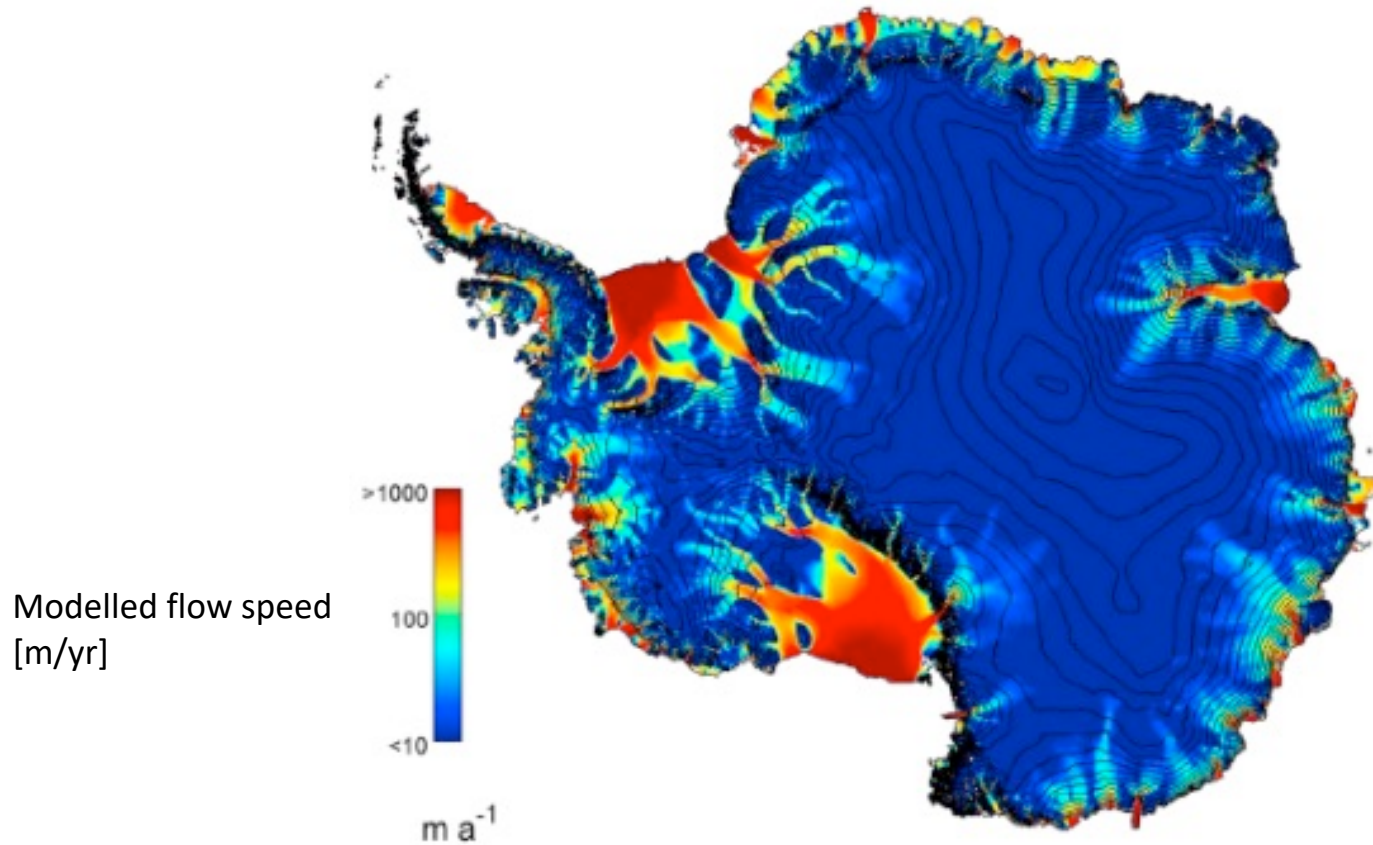
How do we know?

2. Bedforms



How do we know?

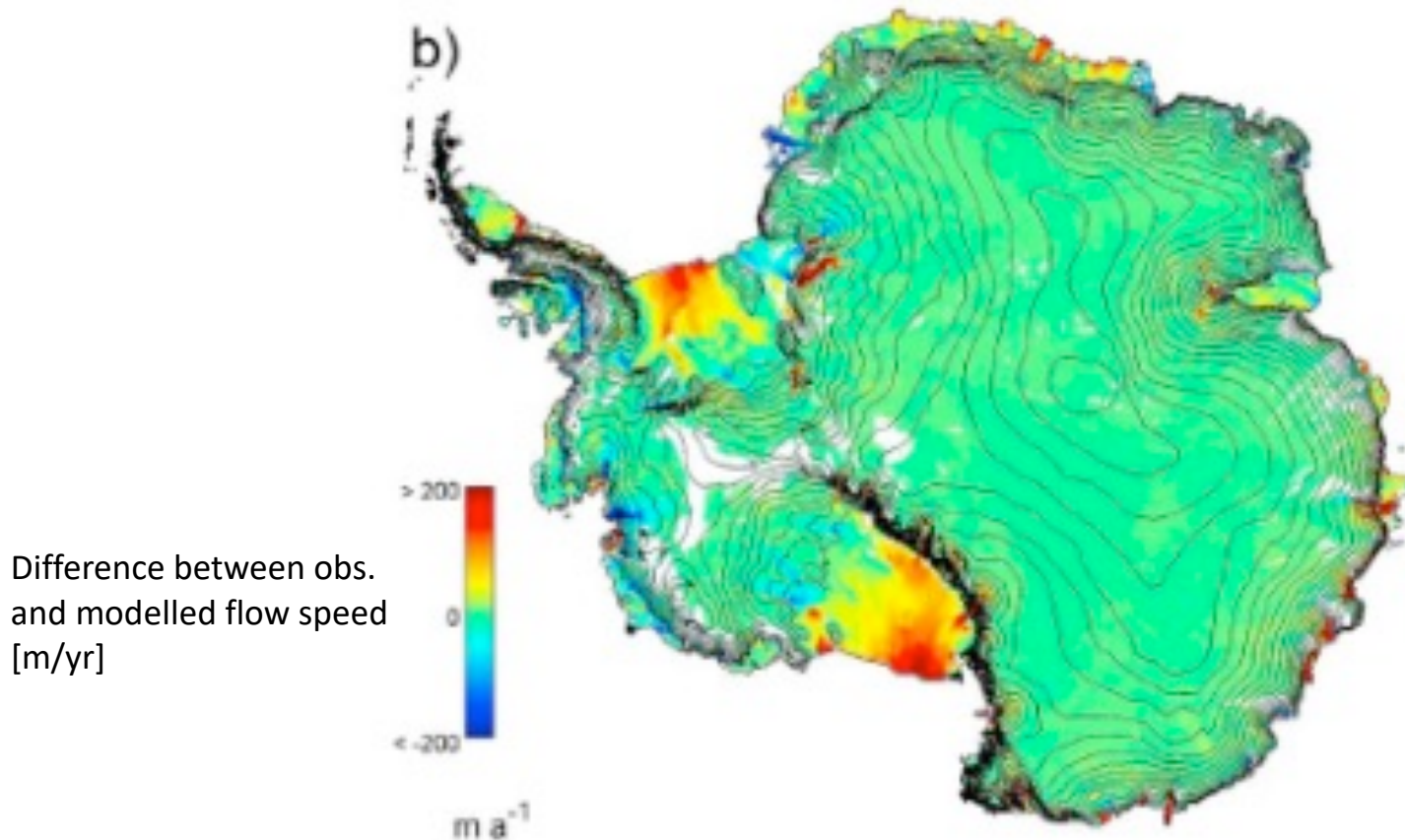
3. Inversions



Arthern et al. (2015)

How do we know?

3. Inversions



Arthern et al. (2015)

How do we know?

3. Inversions

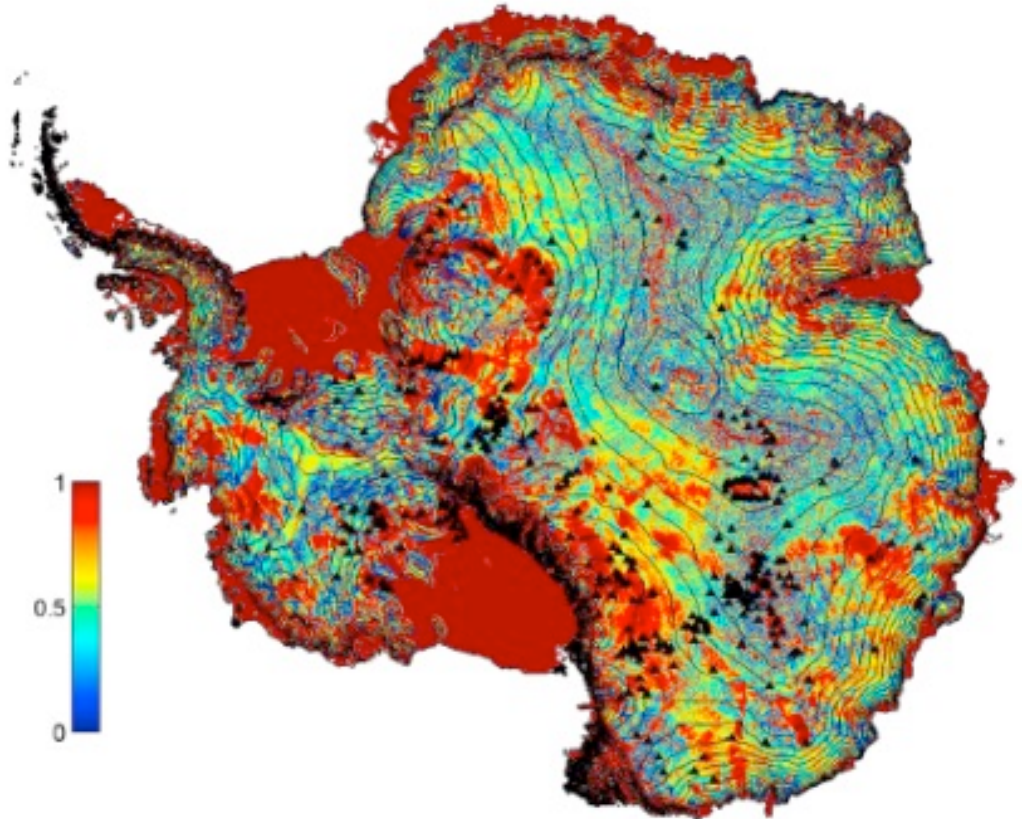


Figure 11. The ratio of flow speed at the bed to flow speed at the surface of the ice sheet.

Arthern et al. (2015)

What controls basal slip?

What controls basal slip?

$$C = \frac{\tau_b}{u_b}$$

What controls basal slip?

$$C = \frac{\tau_b}{u_b}$$

← Approx. equal to driving stress: $\tau_d = \rho g H \frac{\partial H}{\partial x}$

What controls basal slip?

$$C = \frac{\tau_b}{u_b}$$

$$\tau_b = 120 \text{ kPa}$$
$$u_b = 6 \text{ m yr}^{-1}$$

$$C = 20 \frac{\text{kPa}}{\text{m yr}^{-1}}$$

Glacier d' Arolla,
Switzerland



What controls basal slip?

$$C = \frac{\tau_b}{u_b}$$

Variegated Glacier,
Alaska

$$\begin{aligned} \tau_b &= 130 \text{ kPa} \\ u_b &= 90 \text{ m yr}^{-1} \end{aligned} \quad C = 1 \frac{\text{kPa}}{\text{m yr}^{-1}}$$



Lecture 7: Basal Slip
Austin Post, 1965

What controls basal slip?

$$C = \frac{\tau_b}{u_b}$$

$$\begin{array}{l} \tau_b = 130 \text{ kPa} \\ u_b = 90 \text{ m yr}^{-1} \end{array} \quad C = 1 \frac{\text{kPa}}{\text{m yr}^{-1}}$$

Variegated Glacier, Alaska

$$\begin{array}{l} \tau_b = 130 \text{ kPa} \\ u_b = 4800 \text{ m yr}^{-1} \end{array} \quad C = 0.03 \frac{\text{kPa}}{\text{m yr}^{-1}}$$



What controls basal slip?

$$u_b = \frac{\tau_b}{C}$$

Variegated Glacier,
Alaska

$$\begin{aligned} \tau_b &= 130 \text{ kPa} \\ u_b &= 90 \text{ m yr}^{-1} \end{aligned} \quad C = 1 \frac{\text{kPa}}{\text{m yr}^{-1}}$$

$$\begin{aligned} \tau_b &= 130 \text{ kPa} \\ u_b &= 4800 \text{ m yr}^{-1} \end{aligned} \quad C = 0.03 \frac{\text{kPa}}{\text{m yr}^{-1}}$$

Not just τ_b !

Bed properties:
roughness and whether there are sediments present

Hydrology: water pressure, water volume stored

Hard glacier beds



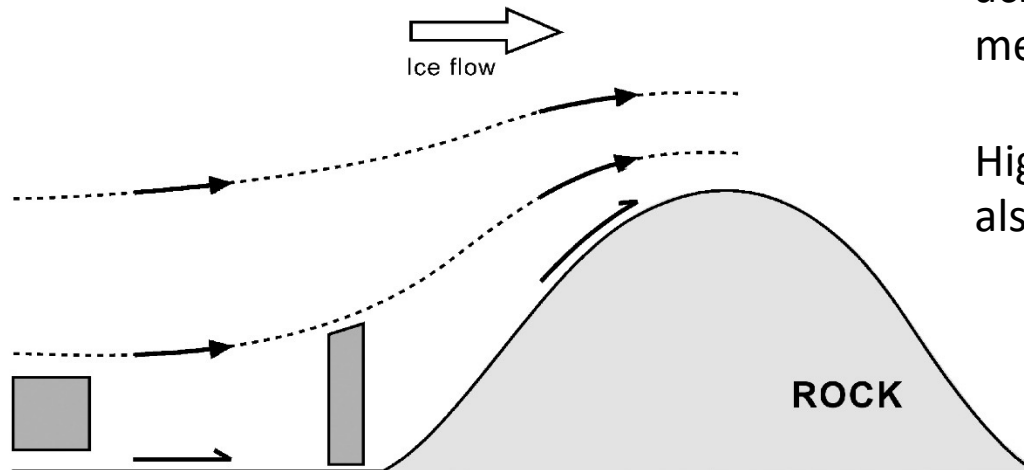
Weertman's theory for hard beds

a

The diagram illustrates a rock dome with a central peak. A large white arrow at the top points to the right, labeled "Ice flow". A curved black arrow on the peak points to the right, labeled "Water flow". A straight black arrow at the base points to the left, labeled "Heat flow". The left side of the dome is labeled "Overpressure" and "Melting point" with a dotted line. The right side is labeled "Underpressure" and "Melting point" with a dotted line. The word "ROCK" is written in large bold letters at the bottom right. The left slope is labeled "Melting" and the right slope is labeled "Freezing".

- Two mechanisms:

b



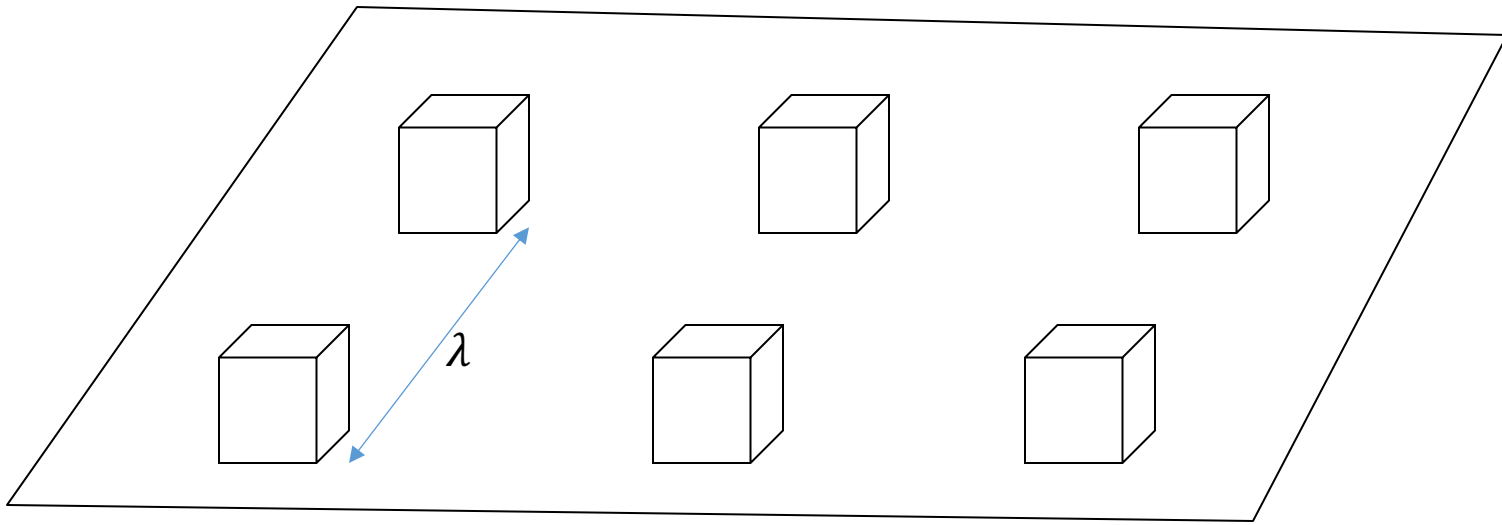
Lecture 7: Basal Slip

Regelation demo:

<https://www.youtube.com/watch?v=qQCVnjGUv24>

Weertman's theory for hard beds

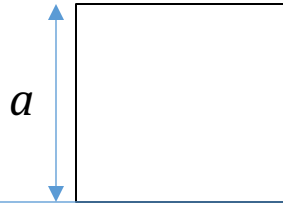
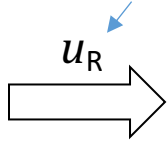
Basal shear stress: τ_b



Force on each cube: $\tau_b \lambda^2$

Weertman's theory for hard beds

Sliding speed due to regelation

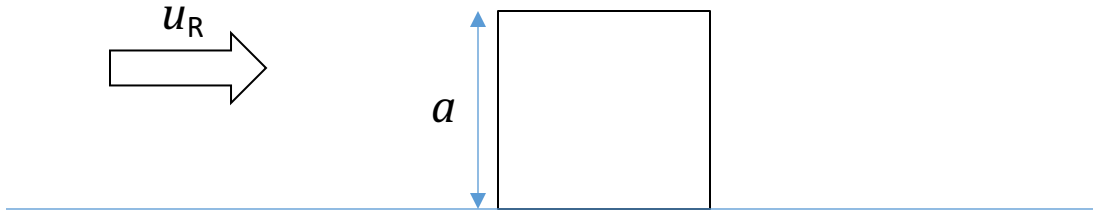


Regelation

$$\text{Force on each cube} = \tau_b \lambda^2$$

Weertman's theory for hard beds

Regelation

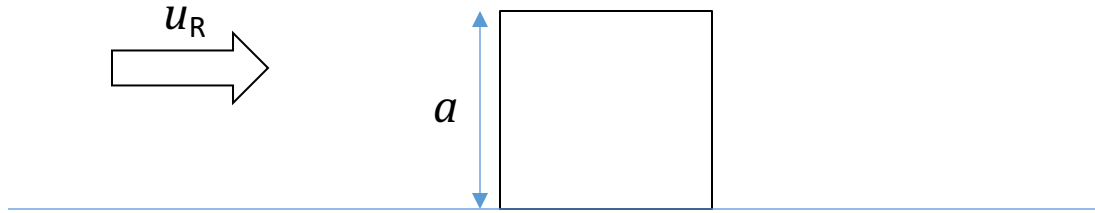


Force on each cube = $\tau_b \lambda^2$

Force on each side of the cube = $\frac{1}{2} \tau_b \lambda^2$

Weertman's theory for hard beds

Regelation



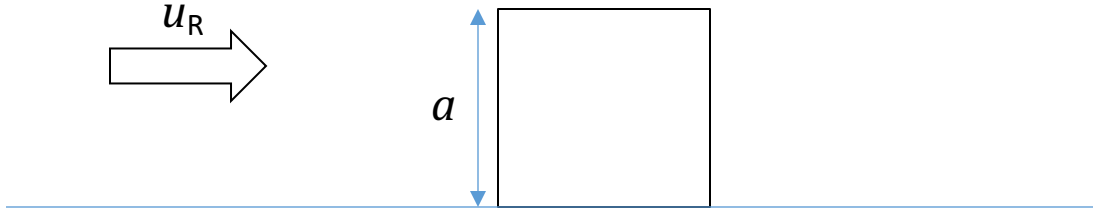
Force on each cube = $\tau_b \lambda^2$

Force on each side of the cube = $\frac{1}{2} \tau_b \lambda^2$

Stress on each side of the cube = $\frac{1}{2} \tau_b \left(\frac{\lambda}{a} \right)^2$

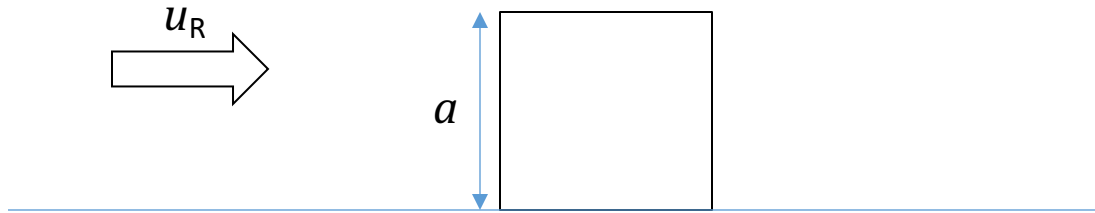
Weertman's theory for hard beds

Regelation



Total differential stress across each cube = $\tau_b \left(\frac{\lambda}{a} \right)^2$

Weertman's theory for hard beds



Regelation

$\mathcal{B}$ is the dependence of the melting point of water on pressure.

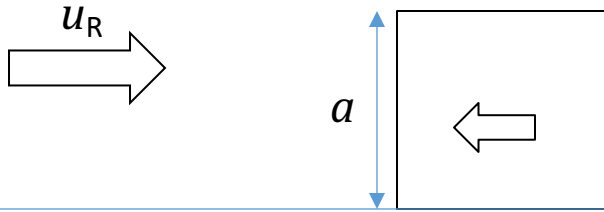
$$= 9.8 \times 10^{-5} \text{ K kPa}^{-1}$$

$$(= 8.7 \times 10^{-4} \text{ K m}^{-1} \text{ in ice})$$

$$\text{Total differential stress across each cube} = \tau_b \left(\frac{\lambda}{a} \right)^2$$

Melting point is a function of stress, so difference in T across the cube is $\delta T = \mathcal{B} \tau_b \left(\frac{\lambda}{a} \right)^2$

Weertman's theory for hard beds



Regelation

k is thermal conductivity of the rock.

...and heat flow through the cube = $ak\mathcal{B}\tau_b \left(\frac{\lambda}{a}\right)^2$

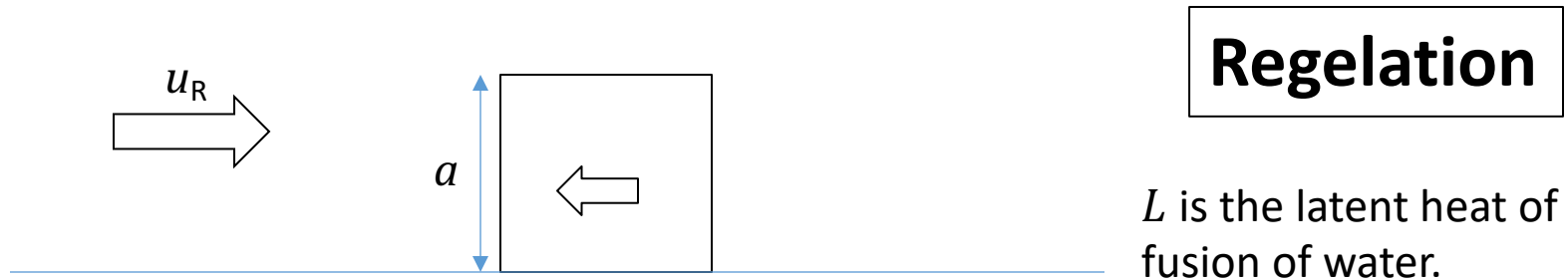
a^2 = Cross-sectional area

Heat flux per unit cross-sectional area, $q = k \frac{dT}{dx}$, so total heat flux is

$$a^2 k \frac{\delta T}{a} = ak\delta T = ak\mathcal{B}\tau_b \left(\frac{\lambda}{a}\right)^2$$

Distance over which temperature gradient acts

Weertman's theory for hard beds



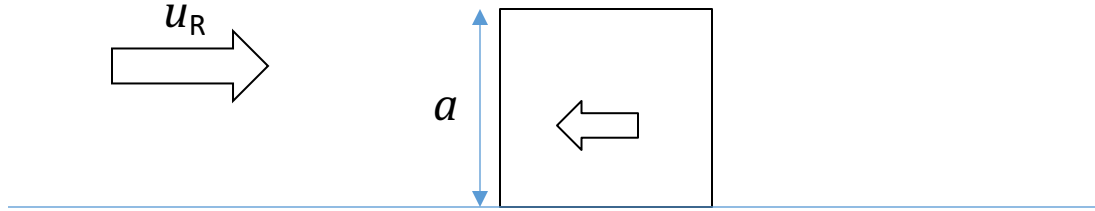
...and heat flow through the cube = $ak\mathcal{B}\tau_b \left(\frac{\lambda}{a}\right)^2$

Volume of ice melted per unit time due to this heat is

$$\dot{m} = \frac{ak\mathcal{B}\tau_b}{\rho_i L} \left(\frac{\lambda}{a}\right)^2$$

Weertman's theory for hard beds

Regelation



...and heat flow through the cube = $ak\mathcal{B}\tau_b \left(\frac{\lambda}{a}\right)^2$

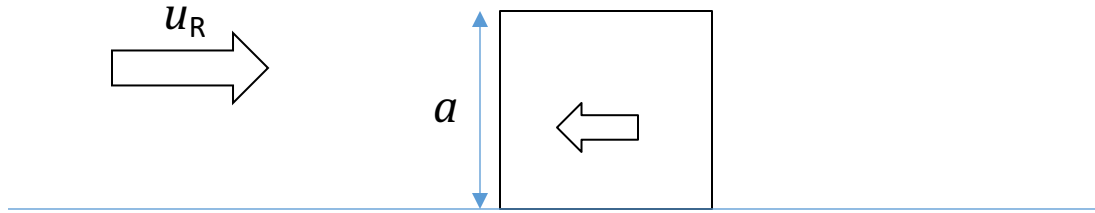
Volume of ice melted per unit time due to this heat is

$$\dot{m} = \frac{ak\mathcal{B}\tau_b}{\rho_i L} \left(\frac{\lambda}{a}\right)^2$$

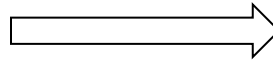
But flow rate is controlled by the melt rate so $\dot{m} = u_R a^2$

Weertman's theory for hard beds

Regelation



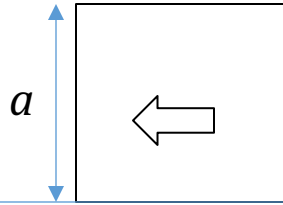
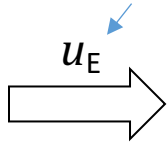
$$u_R a^2 = \frac{a k \mathcal{B} \tau_b}{\rho_i L} \left(\frac{\lambda}{a} \right)^2$$



$$u_R = \frac{k \mathcal{B} \tau_b}{a \rho_i L} \left(\frac{\lambda}{a} \right)^2$$

Weertman's theory for hard beds

Sliding speed due to enhanced creep

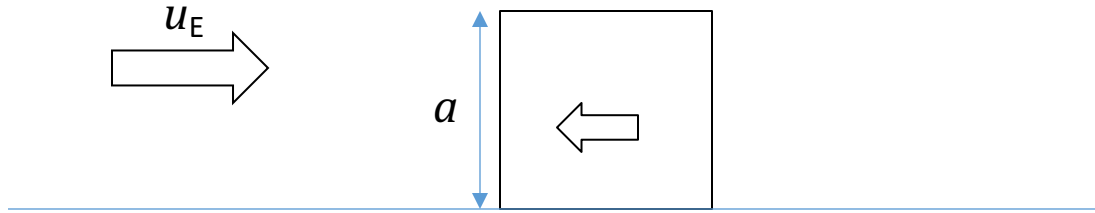


Enhanced Creep

$$\text{Stress on each side of the cube} = \frac{1}{2} \tau_b \left(\frac{\lambda}{a} \right)^2$$

Weertman's theory for hard beds

Enhanced Creep

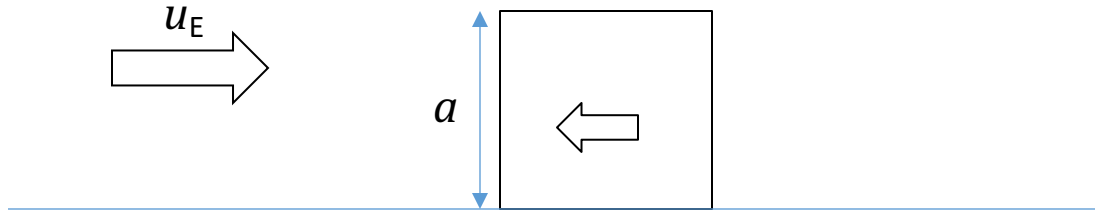


Stress on each side of the cube $= \frac{1}{2} \tau_b \left(\frac{\lambda}{a} \right)^2$

Generates a strain rate $\propto A \left[\tau_b \left(\frac{\lambda}{a} \right)^2 \right]^n$

Weertman's theory for hard beds

Enhanced Creep



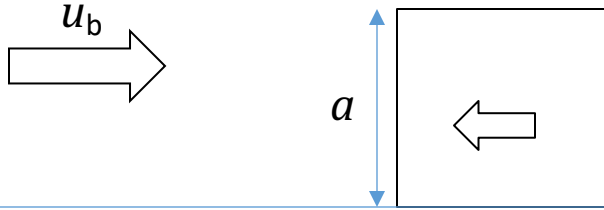
Stress on each side of the cube $= \frac{1}{2} \tau_b \left(\frac{\lambda}{a} \right)^2$

Generates a strain rate $\propto A \left[\tau_b \left(\frac{\lambda}{a} \right)^2 \right]^n$

We assume the strain occurs over a length scale a , so:

$$u_E \propto a A \left[\tau_b \left(\frac{\lambda}{a} \right)^2 \right]^n$$

Weertman's theory for hard beds



Regelation

$$u_R = \frac{k\mathcal{B}\tau_b}{a\rho_i L} \left(\frac{\lambda}{a}\right)^2$$

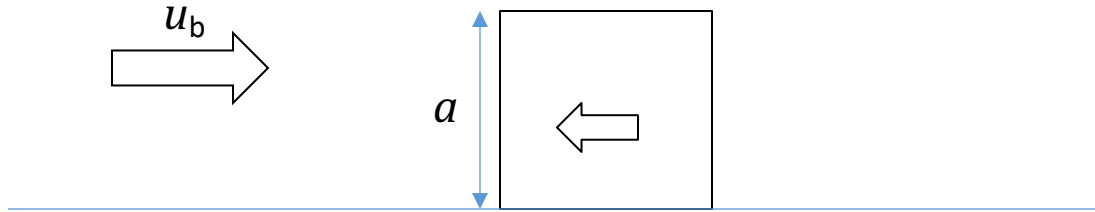
Enhanced Creep

$$u_E \propto aA \left[\tau_b \left(\frac{\lambda}{a}\right)^2 \right]^n$$

For a given obstacle size: $u_b = \max(u_R, u_E)$

But there is a whole range of obstacle sizes,
so one obstacle size controls the sliding speed

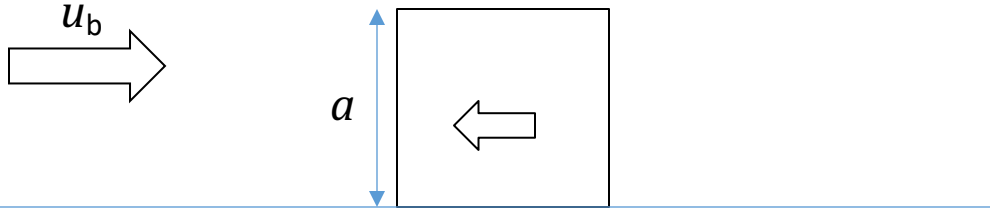
Weertman's theory for hard beds



Equate these and re-arrange for the critical obstacle size, a_c :

$$a^2 \propto \frac{kB\tau_b^{1-n}R^{2(n-1)}}{\rho_i LA}$$

Weertman's theory for hard beds



$$u_R = \frac{k\mathcal{B}\tau_b}{a\rho_i L} \left(\frac{\lambda}{a}\right)^2$$

Equate these and re-arrange for the critical obstacle size, a_c :

$$a^2 \propto \frac{k\mathcal{B}\tau_b^{1-n} R^{2(n-1)}}{\rho_i L A}$$

Substituting back into the regolation equation (above) gives:

$$u_b \propto \left(\frac{\sqrt{\tau_b}}{R}\right)^{n+1}$$

$$R = \frac{\lambda}{a}$$

Assumptions are restrictive and this is a crude model, but it captures some essence of the processes.

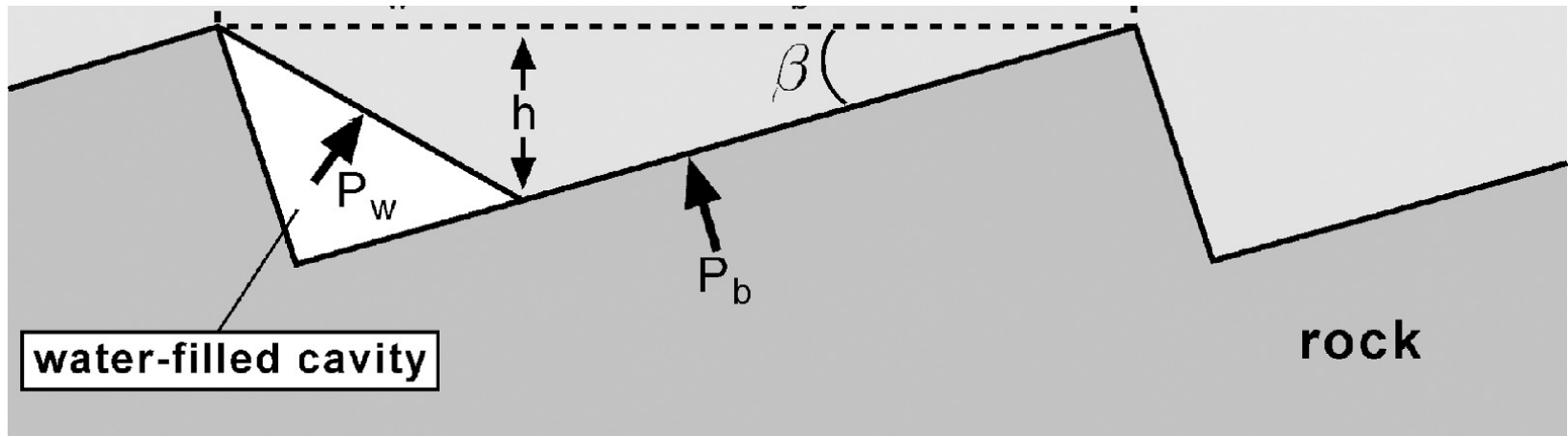
Cavitation

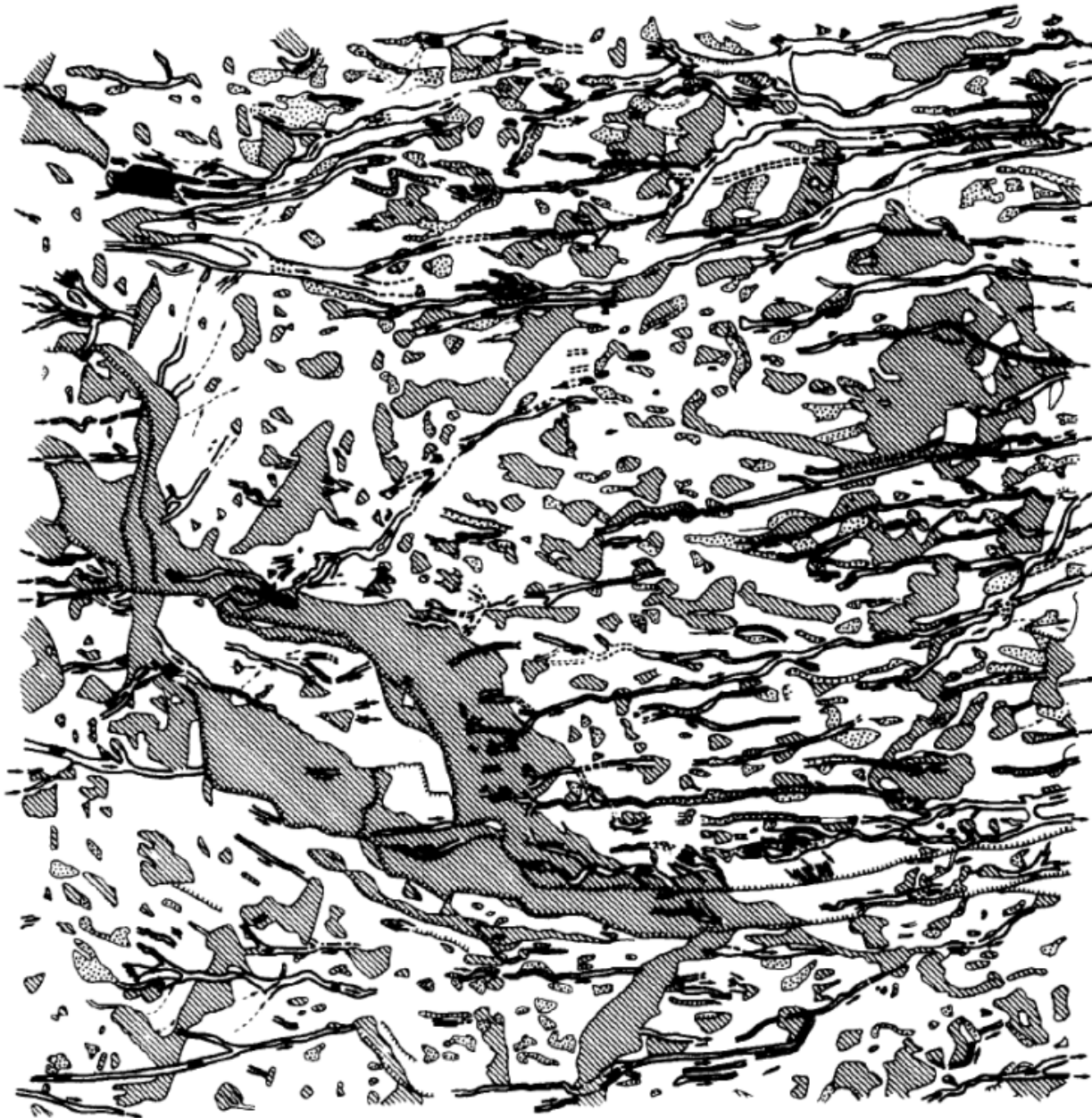
N is the effective pressure = $p_i - p_w$

Where water pressure is high enough, water filled cavities form.

This increases sliding....

$$u_b \propto \frac{\tau_b^p}{RN^q}$$



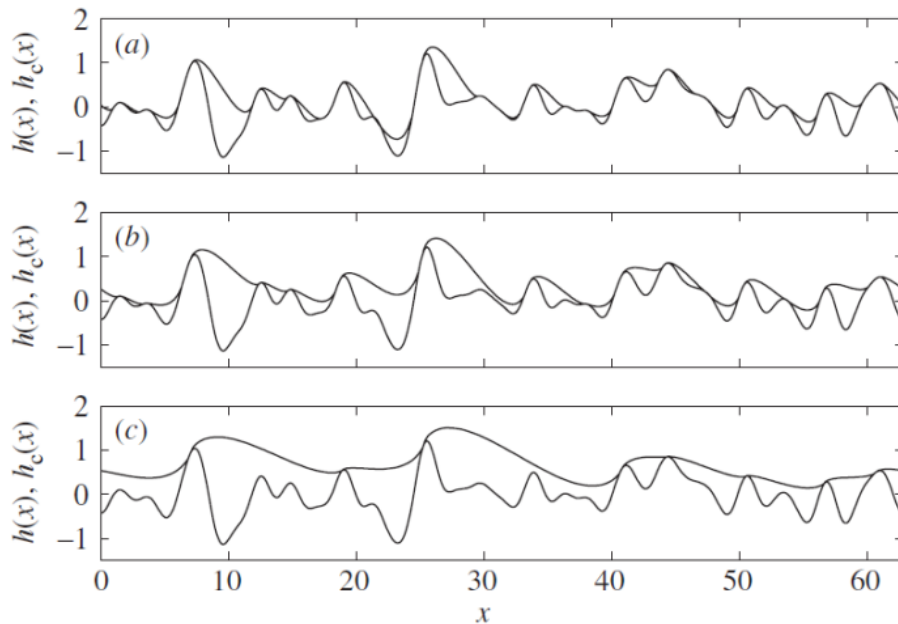


Sharp et al, (1989)

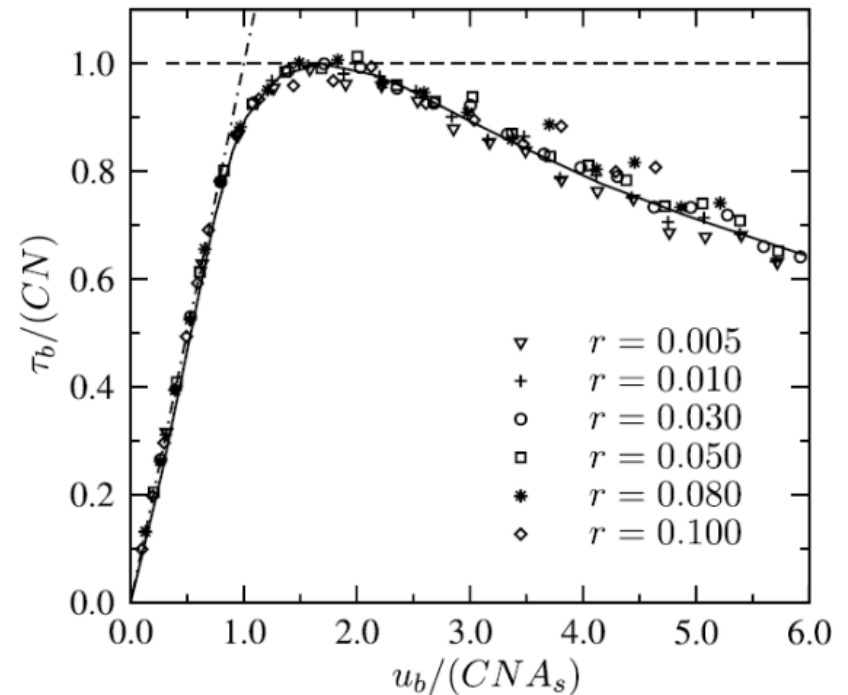
Iken's bound

Iken (1981)

Shear stress generated at the bed cannot exceed some bound because cavities grow so much that they start drowning smaller ones downstream

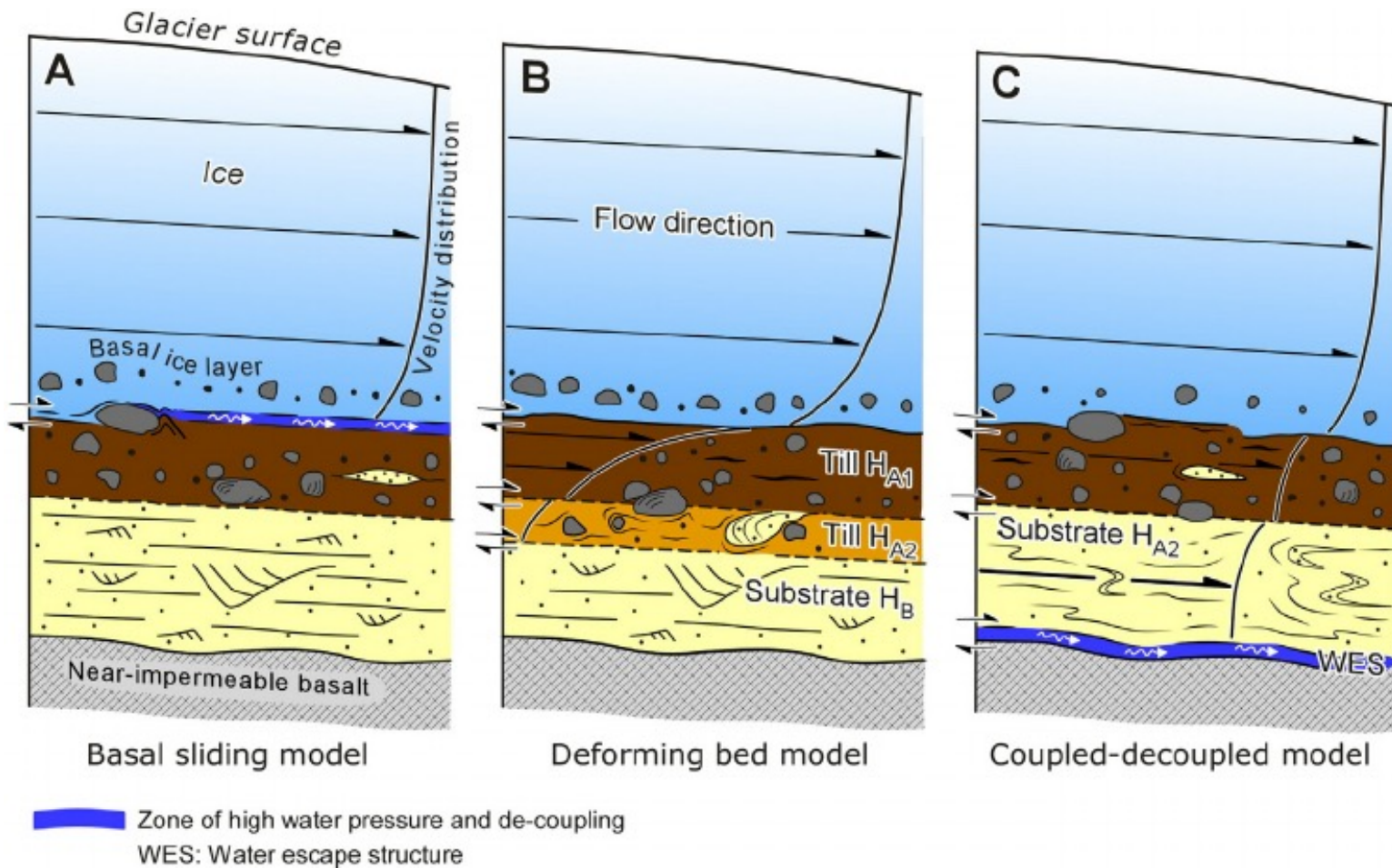


Schoof (2005)



Gagliardini et al. (2007)

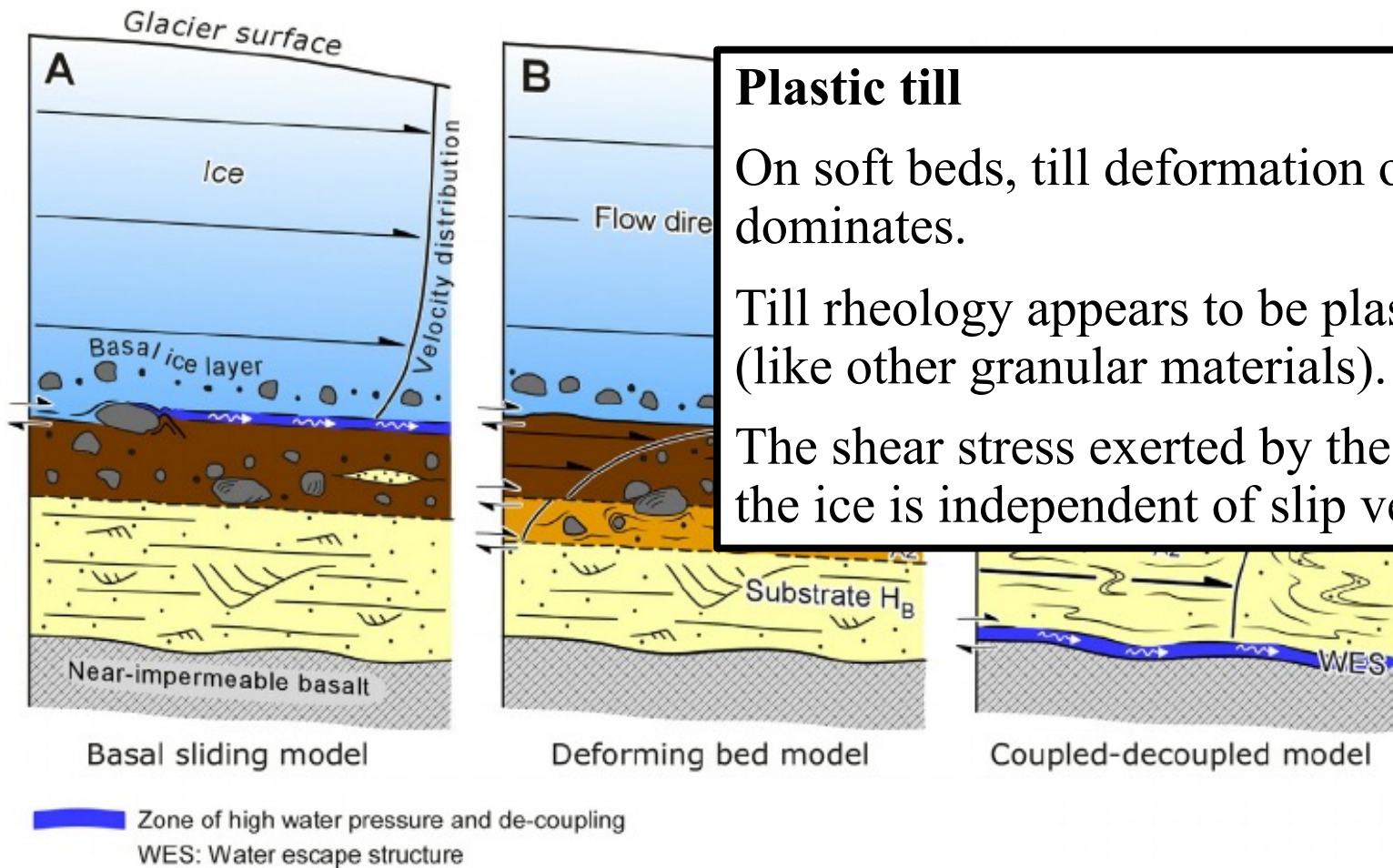
“Soft beds”: deformable material under the ice.



Modified from Kjær et al. (2006)

“Soft beds”:

deformable material under the ice.



Plastic till

On soft beds, till deformation often dominates.

Till rheology appears to be plastic (like other granular materials).

The shear stress exerted by the till on the ice is independent of slip velocity.

Modified from Kjær et al. (2006)

$$\dot{\epsilon} = A\tau^n$$

$$\text{viscosity} \propto \frac{1}{A\tau^{n-1}}$$

What if $n \rightarrow \infty$?

deformation

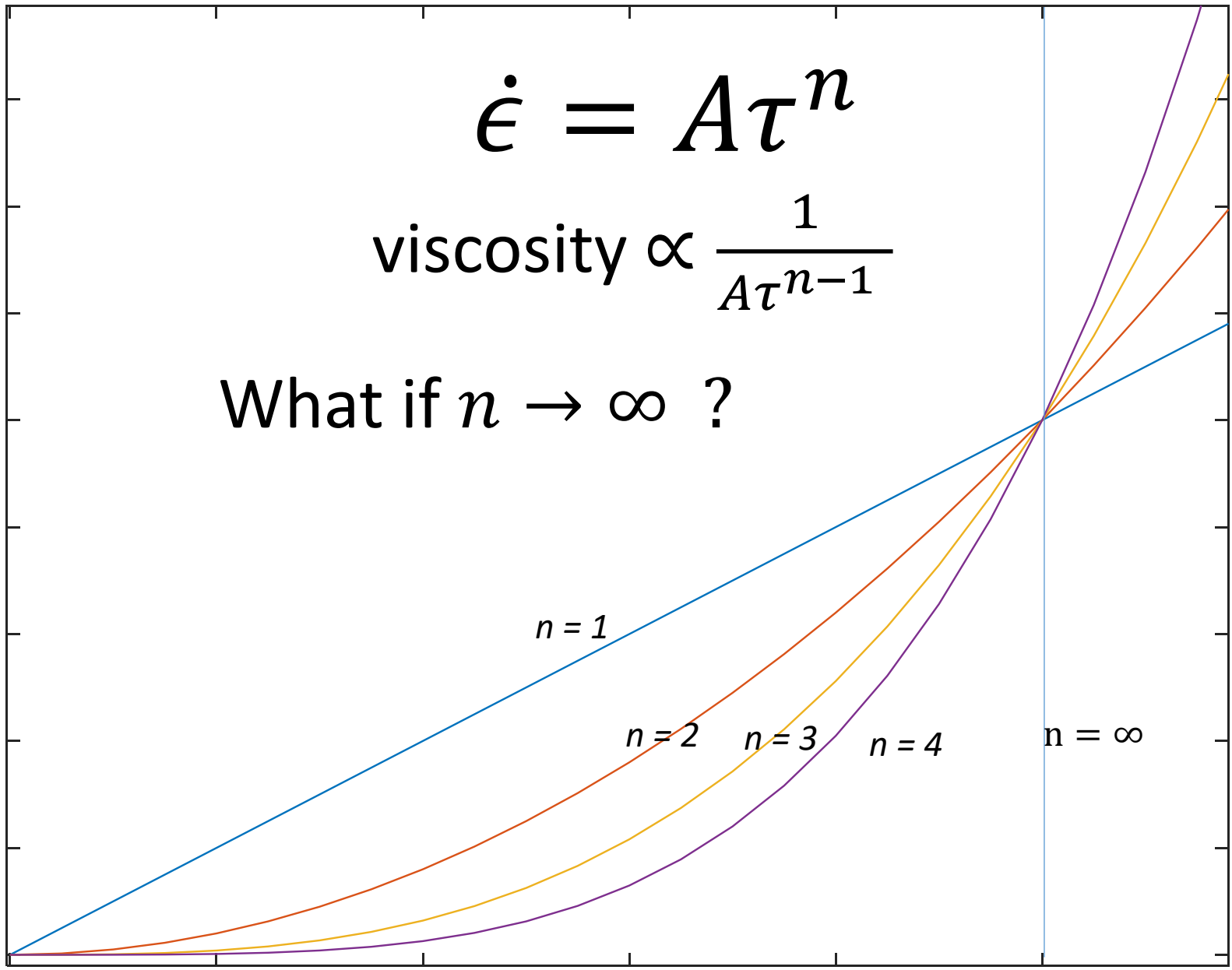
$n = 1$

$n = 2$

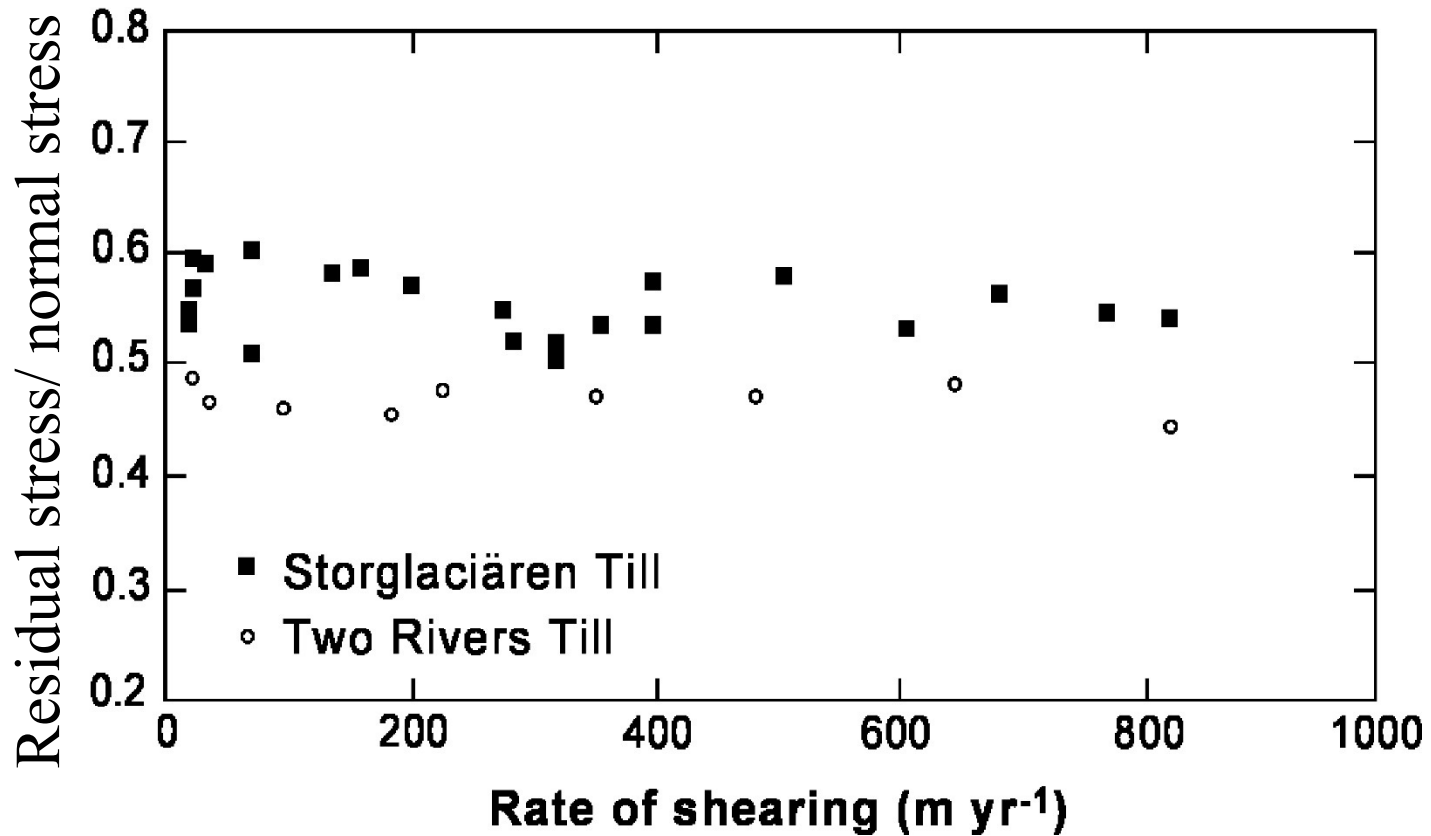
$n = 3$

$n = 4$

$n = \infty$



Mohr-Coulomb sliding



Cuffey and Paterson

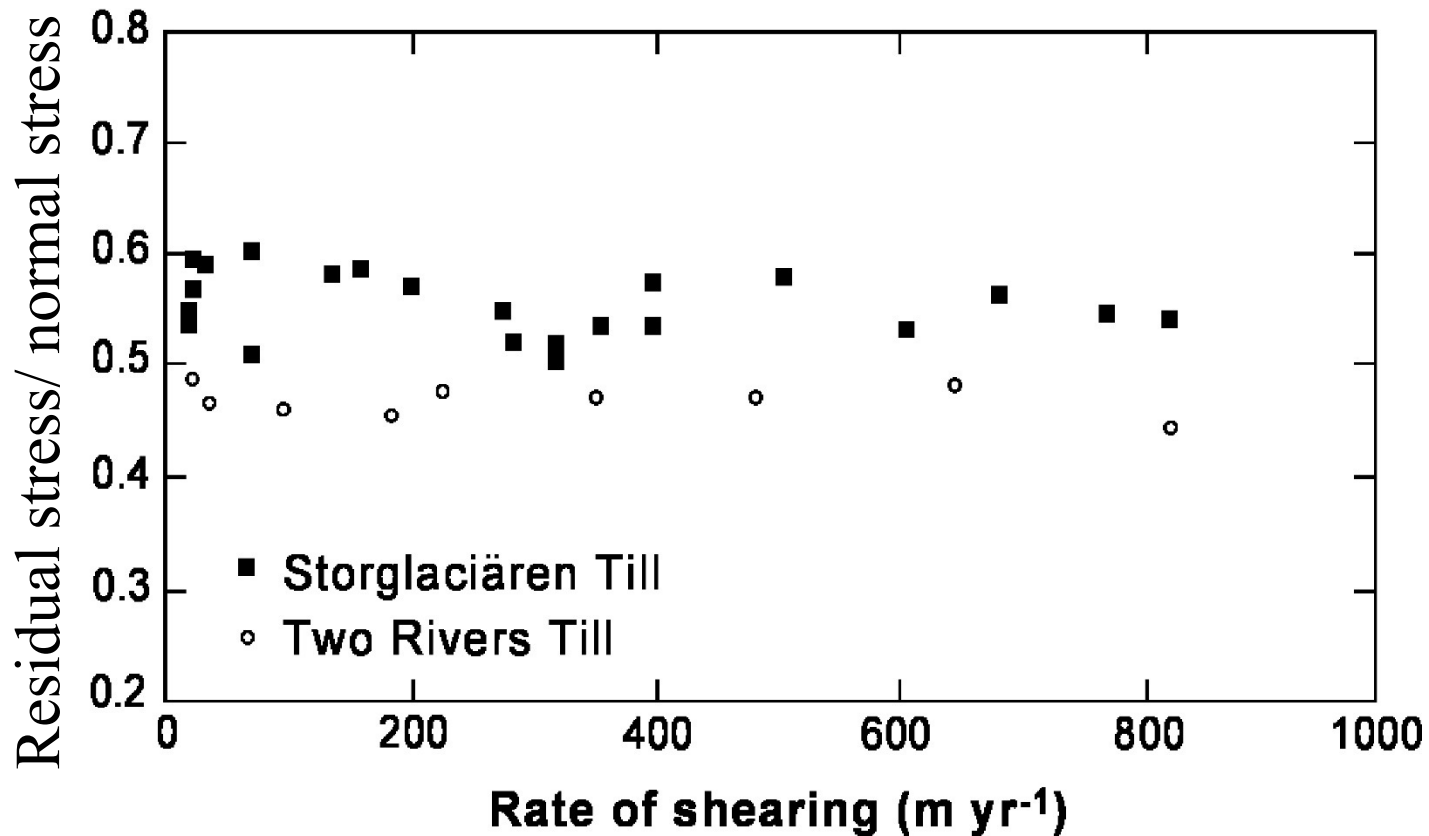
Mohr-Coulomb sliding

Residual stress

$$\tau_* = c_0 + fN$$

Effective pressure

$$N = P_i - P_w$$



Cuffey and Paterson

Mohr-Coulomb sliding

Residual stress

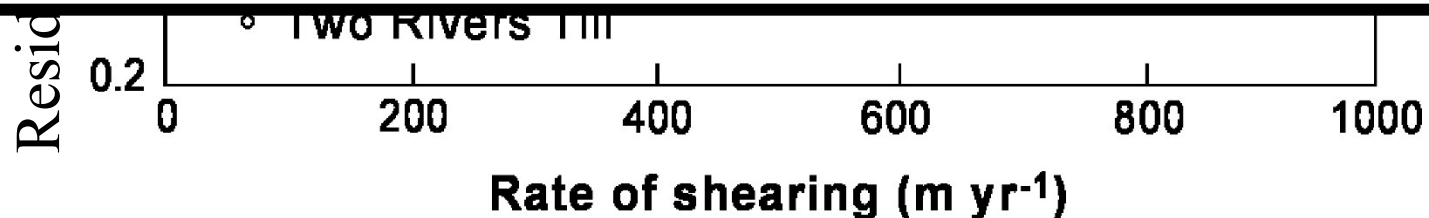
$$\tau_* = c_0 + fN$$

Effective pressure

$$N = P_i - P_w$$

Very different behavior than
“Weertman-style” sliding.

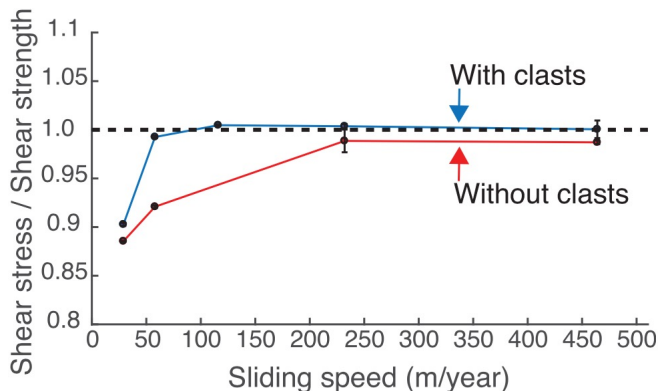
Much current research is aimed at
understanding which of these are
‘correct’.



Cuffey and Paterson

A model to describe both hard beds and soft beds

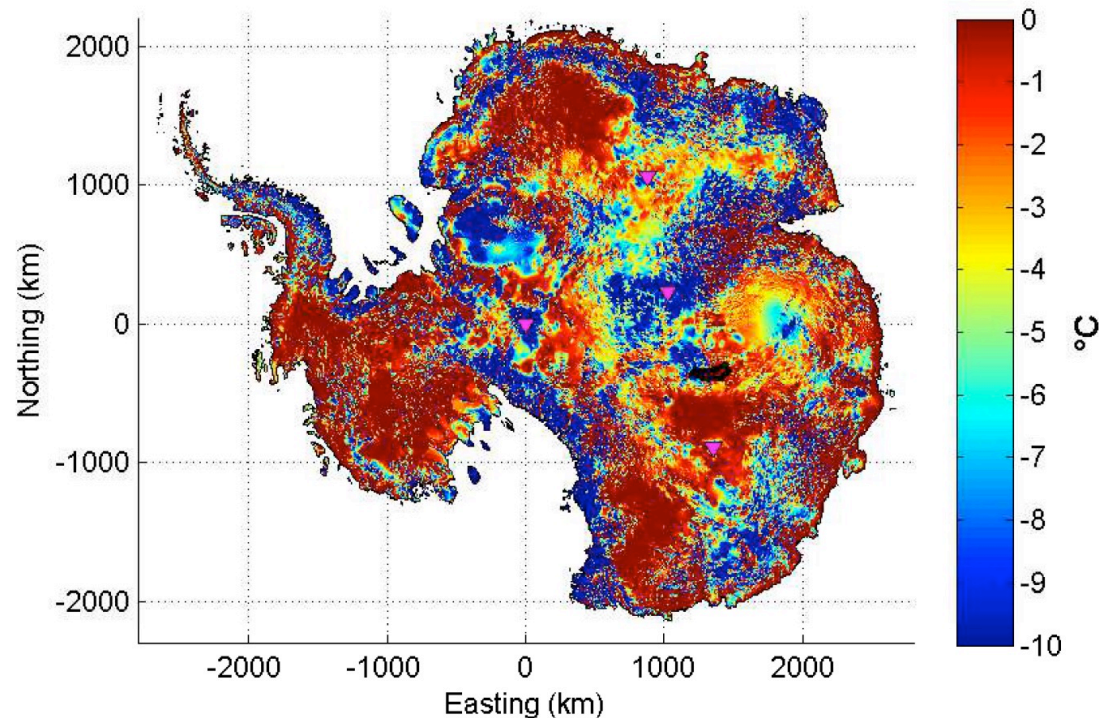
- Zoet and Iverson (2020)
- Experiments with till finds plastic at high speed and power-law at low speed.
- Actually similar to hard beds with cavitation.
- Could be a ‘universal sliding law’



$$\tau_b = N \tan(\phi) \left(\frac{u_b}{u_b + u_t} \right)^{1/p}$$

Implications

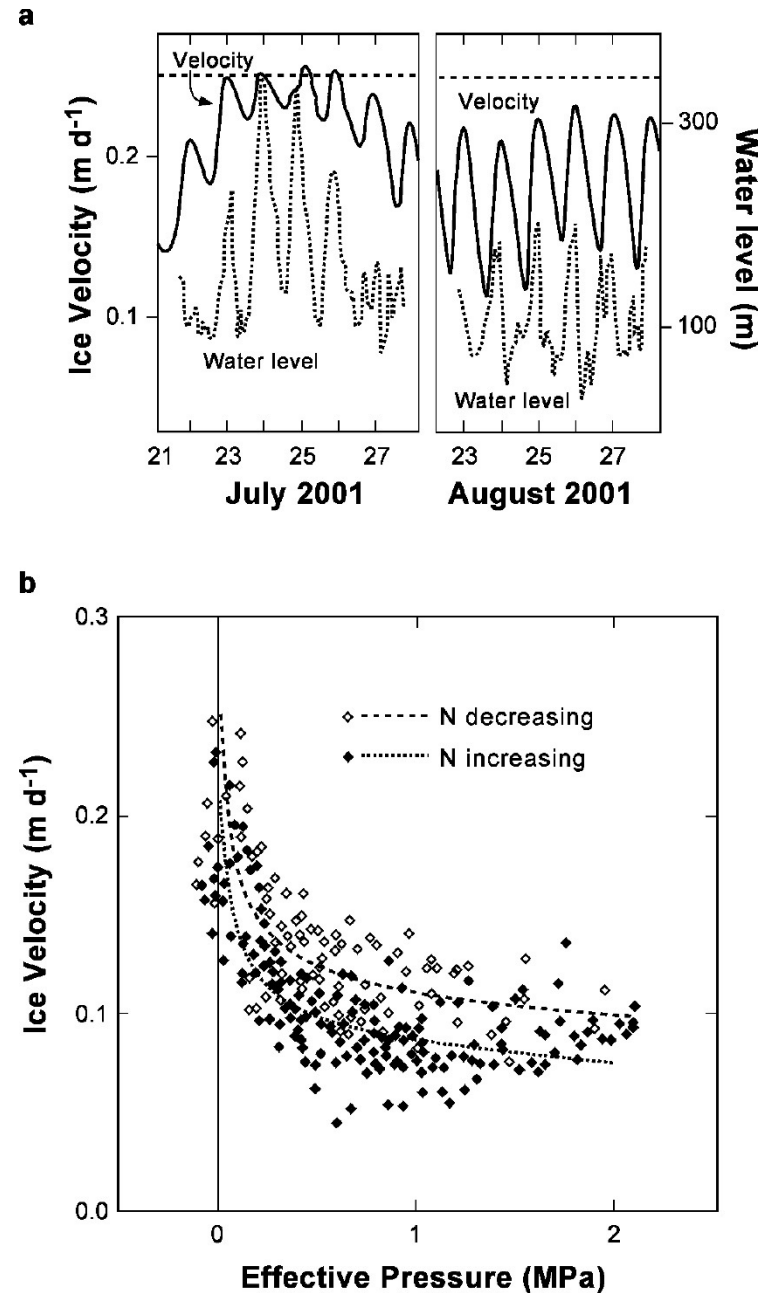
- Temperature at the base controls ice flux by determining if slip occurs.
- Water at the bed controls ice flux by determining how fast slip occurs.
- Glaciers erode and move sediment, modifying the landscape.



Van Liefferinge and Pattyn (2013)

Implications

- Temperature at the base controls ice flux by determining if slip occurs.
- Water at the bed controls ice flux by determining how fast slip occurs.
- Glaciers erode and move sediment, modifying the landscape.



Implications

- Temperature at the base controls ice flux by determining if slip occurs.
- Water at the bed controls ice flux by determining how fast slip occurs.
- Glaciers erode and move sediment, modifying the landscape.



Major questions remain

- Which areas of the bed behave plastically and which behave like Weertman's model? (If these models are even good descriptions of sliding)
- How are water pressures and sliding speed coupled, quantitatively?
- How will the sliding relation evolve in time as ice sheets accelerate and thin?

Summary

- Glaciers slide over their beds.
- Over hard beds, without cavitation, Weertman's sliding model predicts sliding depends on $\tau_b^{\frac{n+1}{2}}$
- Cavitation increases sliding, introduces a dependence on water pressure and in theory can cause unstable sliding.
- Soft beds appear to be plastic, with the resistance provided by the bed not depending on sliding speed.

References

- Arthern, R.J., Hindmarsh, R.C. and Williams, C.R., 2015. Flow speed within the Antarctic ice sheet and its controls inferred from satellite observations. *Journal of Geophysical Research: Earth Surface*, 120(7), pp.1171-1188.
- Blake, W., Fischer, U.H., Bentley, C.R. and Clarke, G.K., 1994. Instruments and Methods: Direct measurement of sliding at the glacier bed. *Journal of Glaciology*, 40(136), pp.595-599.
- Gagliardini, O., Cohen, D., Råback, P. and Zwinger, T., 2007. Finite-element modeling of subglacial cavities and related friction law. *Journal of Geophysical Research: Earth Surface*, 112(F2).
- King, E.C., Hindmarsh, R.C. and Stokes, C.R., 2009. Formation of mega-scale glacial lineations observed beneath a West Antarctic ice stream. *Nature Geoscience*, 2(8), p.585.
- Schoof, C., 2005, March. The effect of cavitation on glacier sliding. In *Proceedings of the Royal Society of London A: Mathematical, Physical and Engineering Sciences* (Vol. 461, No. 2055, pp. 609-627). The Royal Society.
- Sharp, M., Gemmell, J.C. and Tison, J.L., 1989. Structure and stability of the former subglacial drainage system of the Glacier de Tsanfleuron, Switzerland. *Earth Surface Processes and Landforms*, 14(2), pp.119-134.
- Van Liefferinge, B. and Pattyn, F., 2013. Using ice-flow models to evaluate potential sites of million year-old ice in Antarctica. *Climate of the Past*, 9(5), p.2335.